

Capture.png

TShark capturing ICMP traffic sent from the Ubuntu VM to the Kali monitoring system.

In UBUNTU

```
gayathri@gayathri-VirtualBox:~$ ping 192.168.56.101 -c 50
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=0.619 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=1.02 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=0.520 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=0.469 ms
64 bytes from 192.168.56.101: icmp_seq=5 ttl=64 time=1.11 ms
64 bytes from 192.168.56.101: icmp_seq=6 ttl=64 time=0.520 ms
64 bytes from 192.168.56.101: icmp_seq=7 ttl=64 time=0.426 ms
64 bytes from 192.168.56.101: icmp_seq=8 ttl=64 time=0.779 ms
64 bytes from 192.168.56.101: icmp_seq=9 ttl=64 time=0.565 ms
64 bytes from 192.168.56.101: icmp_seq=10 ttl=64 time=0.572 ms
64 bytes from 192.168.56.101: icmp_seq=11 ttl=64 time=0.332 ms
64 bytes from 192.168.56.101: icmp_seq=12 ttl=64 time=1.01 ms
64 bytes from 192.168.56.101: icmp_seq=13 ttl=64 time=0.962 ms
64 bytes from 192.168.56.101: icmp_seq=14 ttl=64 time=0.569 ms
64 bytes from 192.168.56.101: icmp_seq=15 ttl=64 time=0.791 ms
64 bytes from 192.168.56.101: icmp_seq=16 ttl=64 time=0.933 ms
64 bytes from 192.168.56.101: icmp_seq=17 ttl=64 time=1.16 ms
64 bytes from 192.168.56.101: icmp_seq=18 ttl=64 time=1.09 ms
64 bytes from 192.168.56.101: icmp_seq=19 ttl=64 time=1.04 ms
64 bytes from 192.168.56.101: icmp_seq=20 ttl=64 time=0.908 ms
64 bytes from 192.168.56.101: icmp_seq=21 ttl=64 time=0.948 ms
64 bytes from 192.168.56.101: icmp_seq=22 ttl=64 time=1.25 ms
64 bytes from 192.168.56.101: icmp_seq=23 ttl=64 time=0.944 ms
64 bytes from 192.168.56.101: icmp_seq=24 ttl=64 time=0.957 ms
64 bytes from 192.168.56.101: icmp_seq=25 ttl=64 time=0.948 ms
64 bytes from 192.168.56.101: icmp_seq=26 ttl=64 time=0.993 ms
64 bytes from 192.168.56.101: icmp_seq=27 ttl=64 time=0.947 ms
64 bytes from 192.168.56.101: icmp_seq=28 ttl=64 time=1.07 ms
64 bytes from 192.168.56.101: icmp_seq=29 ttl=64 time=1.03 ms
64 bytes from 192.168.56.101: icmp_seq=30 ttl=64 time=0.919 ms
64 bytes from 192.168.56.101: icmp_seq=31 ttl=64 time=0.894 ms
64 bytes from 192.168.56.101: icmp_seq=32 ttl=64 time=0.979 ms
```

```

64 bytes from 192.168.56.101: icmp_seq=33 ttl=64 time=1.09 ms
64 bytes from 192.168.56.101: icmp_seq=34 ttl=64 time=0.839 ms
64 bytes from 192.168.56.101: icmp_seq=35 ttl=64 time=0.926 ms
64 bytes from 192.168.56.101: icmp_seq=36 ttl=64 time=0.730 ms
64 bytes from 192.168.56.101: icmp_seq=37 ttl=64 time=0.705 ms
64 bytes from 192.168.56.101: icmp_seq=38 ttl=64 time=0.878 ms
64 bytes from 192.168.56.101: icmp_seq=39 ttl=64 time=0.864 ms
64 bytes from 192.168.56.101: icmp_seq=40 ttl=64 time=0.566 ms
64 bytes from 192.168.56.101: icmp_seq=41 ttl=64 time=0.803 ms
64 bytes from 192.168.56.101: icmp_seq=42 ttl=64 time=0.721 ms
64 bytes from 192.168.56.101: icmp_seq=43 ttl=64 time=0.851 ms
64 bytes from 192.168.56.101: icmp_seq=44 ttl=64 time=0.773 ms
64 bytes from 192.168.56.101: icmp_seq=45 ttl=64 time=1.01 ms
64 bytes from 192.168.56.101: icmp_seq=46 ttl=64 time=0.896 ms
64 bytes from 192.168.56.101: icmp_seq=47 ttl=64 time=0.890 ms
64 bytes from 192.168.56.101: icmp_seq=48 ttl=64 time=0.961 ms
64 bytes from 192.168.56.101: icmp_seq=49 ttl=64 time=0.852 ms
64 bytes from 192.168.56.101: icmp_seq=50 ttl=64 time=0.932 ms

--- 192.168.56.101 ping statistics ---
50 packets transmitted, 50 received, 0% packet loss, time 49956ms
rtt min/avg/max/mdev = 0.332/0.851/1.245/0.201 ms

```

In Kali

```

(gayathri@kali)-[~]
$ sudo isshark -i eth0
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
1 0.000000000 fe80::c1d:88c6:cd2d:c567 → ff02::1:ff21:647b ICMPv6 86 Neighbor Solicitation for fe80::ded9:16ff:fe21:647b from 0a:00:27:00:00:0d
2 61.030235685 fe80::c1d:88c6:cd2d:c567 → ff02::1:ff21:647b ICMPv6 86 Neighbor Solicitation for fe80::ded9:16ff:fe21:647b from 0a:00:27:00:00:0d
3 61.941437705 PCSSystemtec_fa:e5:8e → Broadcast ARP 60 Who has 192.168.56.102? Tell 192.168.56.100
4 64.748382292 192.168.56.101 → 192.168.56.100 DHCP 324 DHCP Request - Transaction ID 0xb90a6c57
5 64.765321012 PCSSystemtec_fa:e5:8e → Broadcast ARP 60 Who has 192.168.56.101? Tell 192.168.56.100
6 64.765324202 PCSSystemtec_d5:4a:c2 → PCSSystemtec_fa:e5:8e ARP 42 192.168.56.101 is at 08:00:27:d5:4a:c2
7 64.765321414 192.168.56.100 → 192.168.56.101 DHCP 590 DHCP ACK - Transaction ID 0xb90a6c57
8 69.782676368 PCSSystemtec_d5:4a:c2 → PCSSystemtec_fa:e5:8e ARP 42 Who has 192.168.56.100? Tell 192.168.56.101
9 69.782881603 PCSSystemtec_fa:e5:8e → PCSSystemtec_d5:4a:c2 ARP 60 192.168.56.100 is at 08:00:27:fa:e5:8e
10 111.235759614 fe80::a00:27ff:fec7:abd9 → ff02::2 ICMPv6 70 Router Solicitation from 08:00:27:c7:ab:d9
11 122.061403610 fe80::c1d:88c6:cd2d:c567 → ff02::1:ff21:647b ICMPv6 86 Neighbor Solicitation for fe80::ded9:16ff:fe21:647b from 0a:00:27:00:00:0d
12 184.072752759 192.168.56.102 → 192.168.56.101 ICMP 98 Echo (ping) request id=0x10f7, seq=1/256, ttl=64
13 184.072782325 192.168.56.101 → 192.168.56.102 ICMP 98 Echo (ping) reply id=0x10f7, seq=1/256, ttl=64 (request in 12)
14 185.074812031 192.168.56.102 → 192.168.56.101 ICMP 98 Echo (ping) request id=0x10f7, seq=2/512, ttl=64
15 185.074850120 192.168.56.101 → 192.168.56.102 ICMP 98 Echo (ping) reply id=0x10f7, seq=2/512, ttl=64 (request in 14)
16 185.364622163 192.168.56.102 → 224.0.0.251 MDNS 87 Standard query 0x0000 PTR _ipp._tcp.local, "QM" question PTR _ipps._tcp.local, "QM" question
17 186.075620265 192.168.56.102 → 192.168.56.101 ICMP 98 Echo (ping) request id=0x10f7, seq=3/768, ttl=64
18 186.075642942 192.168.56.101 → 192.168.56.102 ICMP 98 Echo (ping) reply id=0x10f7, seq=3/768, ttl=64 (request in 17)
19 186.260201253 fe80::a00:27ff:fec7:abd9 → ff02::fb MDNS 107 Standard query 0x0000 PTR _ipp._tcp.local, "QM" question PTR _ipps._tcp.local, "QM" question
20 187.139621033 192.168.56.102 → 192.168.56.101 ICMP 98 Echo (ping) request id=0x10f7, seq=4/1024, ttl=64
21 187.139642450 192.168.56.101 → 192.168.56.102 ICMP 98 Echo (ping) reply id=0x10f7, seq=4/1024, ttl=64 (request in 20)
22 188.211107657 192.168.56.102 → 192.168.56.101 ICMP 98 Echo (ping) request id=0x10f7, seq=5/1280, ttl=64
23 188.211129473 192.168.56.101 → 192.168.56.102 ICMP 98 Echo (ping) reply id=0x10f7, seq=5/1280, ttl=64 (request in 22)
24 189.079015715 PCSSystemtec_d5:4a:c2 → PCSSystemtec_c7:ab:d9 ARP 42 Who has 192.168.56.102? Tell 192.168.56.101
25 189.079512038 PCSSystemtec_c7:ab:d9 → PCSSystemtec_d5:4a:c2 ARP 60 192.168.56.102 is at 08:00:27:c7:ab:d9
26 189.213438525 192.168.56.102 → 192.168.56.101 ICMP 98 Echo (ping) request id=0x10f7, seq=6/1536, ttl=64
27 189.213484599 192.168.56.101 → 192.168.56.102 ICMP 98 Echo (ping) reply id=0x10f7, seq=6/1536, ttl=64 (request in 26)
28 189.647610487 PCSSystemtec_c7:ab:d9 → PCSSystemtec_d5:4a:c2 ARP 60 Who has 192.168.56.101? Tell 192.168.56.102
29 189.647628618 PCSSystemtec_d5:4a:c2 → PCSSystemtec_c7:ab:d9 ARP 42 192.168.56.101 is at 08:00:27:d5:4a:c2
30 190.275501353 192.168.56.102 → 192.168.56.101 ICMP 98 Echo (ping) request id=0x10f7, seq=7/1792, ttl=64
31 190.275522174 192.168.56.101 → 192.168.56.102 ICMP 98 Echo (ping) reply id=0x10f7, seq=7/1792, ttl=64 (request in 30)
32 191.299570569 192.168.56.102 → 192.168.56.101 ICMP 98 Echo (ping) request id=0x10f7, seq=8/2048, ttl=64
33 191.299609302 192.168.56.101 → 192.168.56.102 ICMP 98 Echo (ping) reply id=0x10f7, seq=8/2048, ttl=64 (request in 32)
34 192.323600372 192.168.56.102 → 192.168.56.101 ICMP 98 Echo (ping) request id=0x10f7, seq=9/2304, ttl=64
35 192.323626797 192.168.56.101 → 192.168.56.102 ICMP 98 Echo (ping) reply id=0x10f7, seq=9/2304, ttl=64 (request in 34)
36 193.349886355 192.168.56.102 → 192.168.56.101 ICMP 98 Echo (ping) request id=0x10f7, seq=10/2560, ttl=64
37 193.349915009 192.168.56.101 → 192.168.56.102 ICMP 98 Echo (ping) reply id=0x10f7, seq=10/2560, ttl=64 (request in 36)
38 194.370533110 192.168.56.102 → 192.168.56.101 ICMP 98 Echo (ping) request id=0x10f7, seq=11/2816, ttl=64
39 194.370556719 192.168.56.101 → 192.168.56.102 ICMP 98 Echo (ping) reply id=0x10f7, seq=11/2816, ttl=64 (request in 38)
40 195.396553462 192.168.56.102 → 192.168.56.101 ICMP 98 Echo (ping) request id=0x10f7, seq=12/3072, ttl=64

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107 227.983913739 192.168.56.102 → 192.168.56.101 ICMP 98 Echo (ping) request id=0x10f7, seq=44/11264, ttl=64
108 227.983946237 192.168.56.101 → 192.168.56.102 ICMP 98 Echo (ping) reply id=0x10f7, seq=44/11264, ttl=64 (request in 107)
109 228.995346352 192.168.56.102 → 192.168.56.101 ICMP 98 Echo (ping) request id=0x10f7, seq=45/11520, ttl=64
110 228.995381968 192.168.56.101 → 192.168.56.102 ICMP 98 Echo (ping) reply id=0x10f7, seq=45/11520, ttl=64 (request in 109)
111 229.997103523 192.168.56.102 → 192.168.56.101 ICMP 98 Echo (ping) request id=0x10f7, seq=46/11776, ttl=64
112 229.997139335 192.168.56.101 → 192.168.56.102 ICMP 98 Echo (ping) reply id=0x10f7, seq=46/11776, ttl=64 (request in 111)
113 231.044763012 192.168.56.102 → 192.168.56.101 ICMP 98 Echo (ping) request id=0x10f7, seq=47/12032, ttl=64
114 231.044800641 192.168.56.101 → 192.168.56.102 ICMP 98 Echo (ping) reply id=0x10f7, seq=47/12032, ttl=64 (request in 113)
115 232.046934603 192.168.56.102 → 192.168.56.101 ICMP 98 Echo (ping) request id=0x10f7, seq=48/12288, ttl=64
116 232.046973478 192.168.56.101 → 192.168.56.102 ICMP 98 Echo (ping) reply id=0x10f7, seq=48/12288, ttl=64 (request in 115)
117 233.048411398 192.168.56.102 → 192.168.56.101 ICMP 98 Echo (ping) request id=0x10f7, seq=49/12544, ttl=64
118 233.048447900 192.168.56.101 → 192.168.56.102 ICMP 98 Echo (ping) reply id=0x10f7, seq=49/12544, ttl=64 (request in 117)
119 234.054255680 192.168.56.102 → 192.168.56.101 ICMP 98 Echo (ping) request id=0x10f7, seq=50/12800, ttl=64
120 234.054293667 192.168.56.101 → 192.168.56.102 ICMP 98 Echo (ping) reply id=0x10f7, seq=50/12800, ttl=64 (request in 119)
121 236.164061265 PCSSystemtec_c7:ab:c2 → PCSSystemtec_d5:4a:c2 ARP 60 Who has 192.168.56.101? Tell 192.168.56.102
122 236.164084078 PCSSystemtec_d5:4a:c2 → PCSSystemtec_c7:ab:c2 ARP 62 192.168.56.101 is at 08:00:27:d5:4a:c2
123 250.128542462 fe80::c1d:88c6:cd2d:c567 → ff02::1:ff21:647b ICMPv6 86 Neighbor Solicitation for fe80::ded9:16ff:fe21:647b from 0a:00:27:00:00:0d
124 314.457516666 192.168.56.1 → 224.0.0.22 IGMPv3 60 Membership Report / Leave group 224.0.0.252
125 314.457517299 fe80::c1d:88c6:cd2d:c567 → ff02::16 ICMPv6 90 Multicast Listener Report Message v2
126 314.460961599 fe80::c1d:88c6:cd2d:c567 → ff02::16 ICMPv6 90 Multicast Listener Report Message v2
127 314.460962088 192.168.56.1 → 224.0.0.22 IGMPv3 60 Membership Report / Join group 224.0.0.252 for any sources
128 314.461665499 192.168.56.1 → 224.0.0.251 MDNS 80 Standard query 0x0000 ANY LAPTOP-47VU47M.local, "QM" question
129 314.462477013 fe80::c1d:88c6:cd2d:c567 → ff02::fb MDNS 100 Standard query 0x0000 ANY LAPTOP-47VU47M.local, "QM" question
130 314.463653767 192.168.56.1 → 224.0.0.251 MDNS 80 Standard query 0x0000 ANY LAPTOP-47VU47M.local, "QM" question
131 314.464064263 fe80::c1d:88c6:cd2d:c567 → ff02::fb MDNS 100 Standard query 0x0000 ANY LAPTOP-47VU47M.local, "QM" question
132 314.464580514 fe80::c1d:88c6:cd2d:c567 → ff02::1:3 LLMNR 94 Standard query 0xdb9e ANY LAPTOP-47VU47M
133 314.464648113 192.168.56.1 → 224.0.0.252 LLMNR 74 Standard query 0xdb9e ANY LAPTOP-47VU47M
134 314.465836465 192.168.56.1 → 224.0.0.251 MDNS 118 Standard query response 0x0000 AAAA fe80::c1d:88c6:cd2d:c567 A 192.168.56.1
135 314.466021042 fe80::c1d:88c6:cd2d:c567 → ff02::fb MDNS 138 Standard query response 0x0000 AAAA fe80::c1d:88c6:cd2d:c567 A 192.168.56.1
136 314.466298541 fe80::c1d:88c6:cd2d:c567 → ff02::fb MDNS 138 Standard query response 0x0000 AAAA fe80::c1d:88c6:cd2d:c567 A 192.168.56.1
137 314.466995163 192.168.56.1 → 224.0.0.251 MDNS 118 Standard query response 0x0000 AAAA fe80::c1d:88c6:cd2d:c567 A 192.168.56.1
138 314.466995470 fe80::c1d:88c6:cd2d:c567 → ff02::16 ICMPv6 90 Multicast Listener Report Message v2
139 314.466944412 192.168.56.1 → 224.0.0.22 IGMPv3 60 Membership Report / Leave group 224.0.0.252
140 314.466968779 fe80::c1d:88c6:cd2d:c567 → ff02::16 ICMPv6 90 Multicast Listener Report Message v2
141 314.467123184 192.168.56.1 → 224.0.0.22 IGMPv3 60 Membership Report / Join group 224.0.0.252 for any sources
142 314.467466277 192.168.56.1 → 224.0.0.251 MDNS 80 Standard query 0x0000 ANY LAPTOP-47VU47M.local, "QM" question
143 314.467635393 fe80::c1d:88c6:cd2d:c567 → ff02::fb MDNS 100 Standard query 0x0000 ANY LAPTOP-47VU47M.local, "QM" question
144 314.467887247 192.168.56.1 → 224.0.0.251 MDNS 118 Standard query response 0x0000 AAAA fe80::c1d:88c6:cd2d:c567 A 192.168.56.1
145 314.467961308 fe80::c1d:88c6:cd2d:c567 → ff02::fb MDNS 138 Standard query response 0x0000 AAAA fe80::c1d:88c6:cd2d:c567 A 192.168.56.1
146 314.468474916 192.168.56.1 → 224.0.0.251 MDNS 80 Standard query 0x0000 ANY LAPTOP-47VU47M.local, "QM" question
147 314.468775118 fe80::c1d:88c6:cd2d:c567 → ff02::fb MDNS 100 Standard query 0x0000 ANY LAPTOP-47VU47M.local, "QM" question
148 314.469030862 192.168.56.1 → 224.0.0.251 MDNS 118 Standard query response 0x0000 AAAA fe80::c1d:88c6:cd2d:c567 A 192.168.56.1
149 314.469187429 fe80::c1d:88c6:cd2d:c567 → ff02::fb MDNS 138 Standard query response 0x0000 AAAA fe80::c1d:88c6:cd2d:c567 A 192.168.56.1
150 314.469187525 fe80::c1d:88c6:cd2d:c567 → ff02::1:3 LLMNR 94 Standard query 0x97d7 ANY LAPTOP-47VU47M
151 314.469436194 192.168.56.1 → 224.0.0.252 LLMNR 74 Standard query 0x97d7 ANY LAPTOP-47VU47M
152 314.452657640 192.168.56.1 → 224.0.0.22 IGMPv3 60 Membership Report / Join group 224.0.0.252 for any sources
153 314.452658242 fe80::c1d:88c6:cd2d:c567 → ff02::16 ICMPv6 90 Multicast Listener Report Message v2
154 319.152415251 fe80::c1d:88c6:cd2d:c567 → ff02::1:ff21:647b ICMPv6 86 Neighbor Solicitation for fe80::ded9:16ff:fe21:647b from 0a:00:27:00:00:0d

```

Output.png

Python analyzer execution showing traffic analysis, suspicious activity detection, and alert generation.

```

(gayathri@kali)-[~]
$ python3 Junior_SOC_Analyst.py
[+] Capturing on eth0 for 100s
Capturing on 'eth0'
50
[+] Capture saved to traffic.pcap
[+] CSV created at traffic.csv

[+] Traffic volume per source IP:

192.168.56.102: 50 packets

[!] Suspicious IP detected: 192.168.56.102
[+] Alert JSON written to alert.json
[+] Sending alert to Airia Agent Execution API...

```

Alert.png

Generated security alert in JSON format after detecting network traffic above the configured threshold.

```
(gayathri@kali)-[~]
$ cat alert.json
{
  "alert_id": "SOC-F446BAAA",
  "alert_type": "Suspicious Network Volume",
  "indicator_type": "ip",
  "indicator_value": "192.168.56.102",
  "destination_host": "Internal-server",
  "destination_ip": "192.168.56.101",
  "evidence": {
    "packet_count": 50,
    "time_window_seconds": 100,
    "data_source": "traffic.pcap"
  },
  "analyst_question": "Is this expected activity or suspicious scanning/noise?"
}
```